	Hence, student must provide $Wx/2$ anticlockwise
7	**B** rate of increase of kinetic energy = $F_{net} \times v = mav$ = (250000)(0.90)(5.0) = 1.1 MW Alternatively, output force of train's engine, F, can be found using N2L: $F_{net} = ma$ F – 15000 = 250000(0.90) F = 240000 N output power of the train's engine = Fv = 240000 × 5.0 = 1 200 000 W However, some of this output power is used against resistive force, while the rest accelerates the train and increases the kinetic energy. Hence, rate of increase of kinetic energy = 1 200 000 – 15000(5.0) = 1.1 MW
8	**B**